



National Institute of Technology
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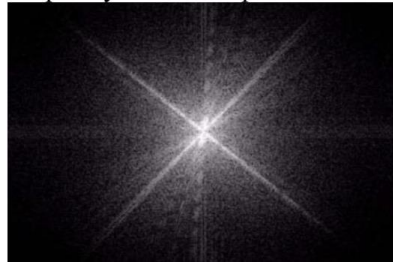
Department	Computer Science and Engineering
Examination	End Semester
Program	B.Tech. 6 th Semester
Session	Spring
Course Code & Course Name	Computer Vision (CS3303)
Marks	50
Duration	180 Minutes

Instructions: 1. In case of any doubt, write your assumptions, write it clearly, and continue.
2. There are 06 questions printed on two pages of the paper. Attempt any five questions.

*CO: Course Outcome; **BL: Bloom's Taxonomy Level.

Q. No.	Questions	Marks	CO*	BL**
1. (A)	When you enter a dark theatre on a bright day, it takes an appreciable interval of time before you can see well enough to find an empty seat. Which of the visual processing concepts takes place in this situation?	[2]		
(B)	Explain the pinhole and perspective projection using a neat diagram. What are the advantages of using lenses instead of pinhole?	[4]	CO1	Remember Understand
(C)	What do you mean by Image Sensing? Explain the Charged Coupled Device (CCD) type of Image Sensors with a neat diagram. Why does each pixel of the CMOS type of image sensor have a smaller light-sensitive area than the CCD type of image sensor?	[4]		
2. (A)	Characterize the three masks by looking at their coefficients. Also, please describe what the individual masks will do to an image f and the applications they usually have. $A = \frac{1}{6} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 0 \end{bmatrix} \quad B = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix} \quad C = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$	[3] (1x3)		
(B)	Which of the following is/are True? If the statement is false, make it accurate without changing its context. i) When a surface coated with cyan pigment is illuminated with white light, no blue light is reflected from the surface. ii) CMY values can be obtained easily from RGB values by subtracting individual RGB values from 1. iii) In the HSI color model, Saturation describes the achromatic notion of intensity.	[3]	CO2	Apply Remember
(C)	What is the goal of bit plane slicing? Why is it useful? State at least one application of it. If a hypothetical gray-scale digital image has 1024 intensity levels, how many bit planes does it have?	[4]		

3. (A) What do you mean by the frequency of an image? Considering the Fourier-transformed image below, construct the filter mask in the frequency domain to produce a smooth and sharpened image. [3]



- (B) Consider an input image of shape $1024 \times 1024 \times 3$. You run this image in a convolutional layer with 32 filters, of kernel size 4×4 with stride 2. How many parameters does this layer have? How can you make the input and output dimensions of the layer same? [3]
- (C) Given the following image. Apply Sobel operators to compute the image gradients. Compute the gradient magnitude and gradient direction. [4]

$$\begin{bmatrix} 10 & 10 & 10 \\ 100 & 100 & 100 \\ 100 & 100 & 100 \end{bmatrix}$$

4. (A) Contrast and compare the two contemporary object detection methods: (i) Region Proposal Based Methods and (ii) Dense Sampling Based Methods. [3]

- (B) Draw the architecture of Faster R-CNN with a neat diagram. Explain its four steps of training. [5]

- (C) Why Faster R-CNN is faster than R-CNN Justify? [2]

5. (A) Why is image blending carried out in image stitching? Discuss two popular image blending methods. [2]

- (B) Explain the line detection algorithm using Hough Transform. Why you should not use generalized Hough Transform instead of Hough Transform to detect lines? [4]

- (C) What is an interest point? How can you extract SIFT interest points? [4]

6. (A) Match the following [2]

- | | |
|---------------|--|
| (i) Dilation | (a) Structured Removal of Image Region Boundary Pixels |
| (ii) Erosion | (b) Shrink Image Regions |
| (iii) Opening | (c) Grow Image Regions |
| (iv) Closing | (d) Structured Filling of Image Region Boundary Pixels |

- (B) Given a Binary Image (A) and Structuring Element (B with centre element is the origin). Compute the opening and closing of A using B. Show the detailed steps. [8] (4+4)

A =

$$\begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

B =

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

CO3

Evaluate
Understand

CO3

Remember
Apply

CO4

Apply
Understand

CO4

Evaluate
Understand

End of Paper